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SUBJECT: Prior Art Report for BevoTag: Wi-Fi Fingerprinting, Inertial Fusion, and Item-Tracking Patents That Inform the Design

INTRODUCTION

This Prior Art Report documents the BevoTag team's search of published research and patents related to the project, and explains how the results have shaped the design. BevoTag is a small tag that locates personal belongings or shared equipment on the University of Texas at Austin campus by using campus Wi-Fi signals and a motion sensor. The team's Project Definition committed to this approach but left four questions open: which Wi-Fi measurement to use, how much surveying a building requires, how to combine Wi-Fi with motion data, and which parts of the design existing patents already cover. To answer these questions before the system design is finalized, the team reviewed five research publications and five U.S. patents. This report first summarizes the design problem and then compares each publication and patent with BevoTag's requirements. Finally, it explains how the findings have changed the design.

OVERVIEW OF THE DESIGN PROBLEM

Students, faculty, and staff often need to find a misplaced item somewhere on campus, yet no existing tracker works reliably indoors without help from other people's phones. Satellite positioning signals, such as those of the Global Positioning System (GPS), do not penetrate buildings well [1]. Consumer tags such as the Apple AirTag and the Samsung Galaxy SmartTag2 update an item's location only when a nearby phone from the same manufacturer detects them [2], [3], so the location of a bag left in an empty lab at night stops updating. Wi-Fi positioning that relies on signal strength alone typically errs by 1 to 5 m [4], enough to report the wrong room. Customers therefore need a tag that reports its own location on any floor, without a passing phone and without new infrastructure such as Bluetooth beacons installed in every room.

BevoTag addresses this need by combining two sources of position information. The tag records the received signal strength indicator (RSSI) of nearby access points and, where its chip supports it, their channel state information (CSI), which describes how each signal changes across its frequency subcarriers. The tag also samples an inertial measurement unit (IMU) that measures acceleration and rotation. A location server compares the Wi-Fi measurements with a survey recorded at known points, a process called fingerprinting, and fuses the result with the IMU data, because an IMU used alone produces a position estimate that drifts by more than 150 m within 60 s [5]. A user application then shows the position on a campus map.

A successful design will identify the room that contains the item in more than 90% of trials, keep the median position error under 5 m, update at least every 10 s while the item moves, and run at least 24 hours on one charge, at a parts cost of no more than $50 per tag. The prototype uses an ESP32-family microcontroller, which exposes CSI to user firmware [6], and the team has already built a data logger and a survey application for three buildings, starting with the Engineering Education and Research Center (EER).

REVIEW OF PRIOR ART

The search was organized around four research questions that the Project Definition left open. Each question corresponds to one design challenge, and a source was kept only if it gave evidence on at least one question:

1. **Wi-Fi measurement:** Does CSI give enough accuracy beyond RSSI to justify its cost, and can a low-cost chip capture it from campus access points?
2. **Survey effort:** How many survey points, samples, and repeated visits does a building need, and how quickly does a survey go out of date?
3. **Sensor fusion:** How should the tag combine Wi-Fi fixes with IMU data to keep drift bounded, and how much does fusion improve accuracy?
4. **Freedom to operate and novelty:** Which parts of the design do existing products and patents already cover, and where can BevoTag differ?

The first subsection reviews research publications that address questions 1 through 3, and the second reviews patents that address all four.

Prior Art Exclusive of Patent Information

Five research publications trace Wi-Fi positioning from the first signal-strength fingerprinting system to CSI capture on the same microcontroller family that BevoTag uses. Two of them establish RSSI and CSI baselines, one tests CSI capture on BevoTag's hardware, one measures the benefit of inertial fusion, and one compares many systems in a realistic, changing environment.

Search Procedure

The team searched IEEE Xplore with keyword queries that combined "Wi-Fi" or "WLAN" (wireless local area network) with "fingerprint," "channel state information," "ESP32," "pedestrian dead reckoning," "inertial," and "Kalman" or "particle filter," and it retrieved each candidate's full text from the publisher or the authors' own pages. The team kept papers that report measured accuracy in real buildings, use hardware comparable in cost to BevoTag's, or evaluate systems in environments that change. Papers were discarded if they relied only on simulation, required infrastructure that the team cannot install, such as ultra-wideband anchors, or addressed a task other than locating a device, unless they measured hardware that BevoTag will use.

RADAR: The First Wi-Fi Fingerprinting System

Bahl and Padmanabhan's RADAR system, presented in 2000, established the two-phase fingerprinting method that BevoTag adopts [7]. In an off-line phase, the authors recorded signal strength from three base stations at 70 known points on a 43.5 m by 22.5 m office floor with more than 50 rooms; in an on-line phase, they located a laptop by finding the survey points whose signal strengths were nearest in "signal space." This empirical method achieved a median error of 2.94 m, which the authors noted is about the size of an office room [7].

RADAR's results inform BevoTag's survey design more than its accuracy targets. The authors found that about 40 survey points and three real-time samples achieved nearly all of the accuracy

of the full data set [7]. When the survey contained only one orientation and the user faced the opposite way, however, the median error rose 67% to 4.90 m, because the user's body changes the signal strength that each base station measures [7]. RADAR's median error is within BevoTag's 5 m ceiling but too coarse for reliable room identification, and its three base stations were dedicated hardware, whereas BevoTag relies on whatever campus access points it hears.

DeepFi: CSI Fingerprinting Compared with RSSI

The clearest evidence on whether CSI improves on signal strength comes from Wang, Gao, Mao, and Pandey's DeepFi system, which compares CSI and RSSI fingerprints measured in the same rooms and was published in *IEEE Transactions on Vehicular Technology* [8]. The authors recorded the amplitudes of 30 subcarriers on each of three antennas of a laptop Wi-Fi card, trained a neural network for each of 50 survey points, and estimated position as a probability-weighted average of the survey points. They compared DeepFi with an earlier CSI method and with Horus, a widely cited probabilistic RSSI system [9], using the same data.

CSI fingerprints were more accurate than RSSI fingerprints in both test rooms. In a mostly empty 4 m by 7 m living room, DeepFi's mean error was 0.94 m and Horus's was 1.54 m; in a cluttered 6 m by 9 m laboratory with few line-of-sight paths, the errors were 1.81 m and 2.60 m [8]. These reductions of 39% and 30% justify evaluating CSI for BevoTag. The study differs from BevoTag in three important ways, however: it used one access point that the authors controlled, a laptop card with three antennas rather than BevoTag's single-antenna chip, and environments much smaller than a typical building floor.

ESP32 CSI Toolkit: CSI Capture on a Microcontroller

Because DeepFi relied on a laptop card, evidence was also needed that BevoTag's own hardware can capture CSI. Hernandez and Bulut showed in 2020 that the ESP32 microcontroller can capture CSI without the laptops and modified drivers that earlier studies required [10]. Their open-source toolkit runs on a module that costs less than $10, weighs less than 10 g, and reports 64 subcarriers from a single antenna [10]. The authors used it for activity recognition rather than localization, but they measured the capture rates that matter to BevoTag.

These capture rates reveal a limit on BevoTag's design. When the ESP32 was connected to the transmitter, it captured CSI from about 600 frames per second; when it only listened, which BevoTag will do to hear many campus access points, it captured just 34.5% as many frames, with greater variation from second to second, because a passive receiver cannot request retransmission [10]. This result confirms that the chosen hardware can supply CSI, but it suggests that the tag will collect far fewer CSI samples from campus access points than from a transmitter it controls.

Evennou and Marx: Fusing Wi-Fi with Dead Reckoning

For the fusion stage, the closest published precedent is a 2006 *EURASIP Journal on Applied Signal Processing* paper by Evennou and Marx of France Telecom, who fused Wi-Fi fingerprinting with pedestrian dead reckoning [11]. A belt-worn box counted steps with an accelerometer and tracked heading with a gyroscope. The authors compared a Kalman filter with a particle filter that uses the building map to eliminate any position hypothesis that crosses a wall.

Each step of filtering and fusion improved accuracy in their 40 m by 40 m office building with four access points. Along a corridor trajectory, nearest-neighbor fingerprinting gave a mean error of 3.32 m, a statistical fingerprint method 2.88 m, a Kalman filter 2.56 m, the map-aware particle filter 1.86 m, and the particle filter with inertial data 1.53 m [11]. The authors also observed that the particle filter's many wall checks make it computationally expensive [11]. These results support fusion and the use of floor plans, which BevoTag's survey application already contains, but they were measured with a laptop doing the processing and a survey of only one point per room.

Microsoft Indoor Localization Competition: Performance Under Change

Each study above tested one system in a space of the authors' choosing. In contrast, Lymberopoulos *et al.* reported how 22 indoor positioning systems performed when tested in the same unfamiliar space at the 2014 Microsoft Indoor Localization Competition [12]. Teams were given seven hours to install hardware or survey an area of about 300 m^2 that spanned two rooms and a hallway, and the organizers then rearranged the furniture in one room before measuring each system at 20 test points.

The competition shows that Wi-Fi fingerprinting can rival custom infrastructure but degrades when a room changes. The best system that relied only on existing access points and phone sensors had a mean error of 1.6 m, better than most systems that installed their own hardware, and half of all teams achieved errors below 3 m [12]. Most teams needed all seven hours to survey or install, and rearranging the furniture raised the error of the leading Wi-Fi systems by up to 0.94 m [12]. Because campus rooms change daily as people move chairs and walk through hallways, these findings indicate that a one-time survey of EER will not stay accurate.

Table 1 compares the five publications with BevoTag. Together, they support fingerprinting with inertial fusion but leave a single-antenna tag in a changing building untested.

Table 1. Research Publications Compared with BevoTag Requirements

Source	Question	Key Finding	Difference from BevoTag
RADAR [7]	2	2.94 m median; survey should cover different orientations	Dedicated base stations; no motion sensing
DeepFi [8]	1	CSI mean error 30–39% below RSSI	One controlled access point; three-antenna laptop card
ESP32 CSI Toolkit [10]	1	Module under $10 captures 64-subcarrier CSI; passive capture rate 34.5% of active	Used for activity sensing, not localization
Evennou and Marx [11]	3	Map-aware particle filter with inertial data: 1.53 m mean error	Laptop processing; one survey point per room
Microsoft competition [12]	2, 3	Best infrastructure-free system 1.6 m; furniture changes add up to 0.94 m	Two rooms and a hallway; phones, not tags

Patent Search and Findings

Five U.S. patents cover the survey database, the CSI fingerprint, the fusion step, and the crowdsourced tag model that BevoTag will either build on or avoid. Each patent's claims are described below and compared with BevoTag's design; these comparisons are engineering judgments, not legal opinions.

Search Procedure

The team searched Google Patents with combinations of "Wi-Fi" or "wireless access point" with "fingerprint," "channel state information," "subcarrier," "dead reckoning," "inertial," "lost and found," and "tag," and it retrieved the full grant documents from the U.S. Patent and Trademark Office. For each candidate, the team read the independent claims and recorded the assignee, grant date, and status, keeping the patents whose claims describe a function that BevoTag performs. Google Patents' status and expiration dates are estimates rather than legal conclusions, so before the design is finalized, the team will ask the Engineering Library staff to help confirm the status of any patent of concern.

US 7,414,988: Server for Updating a Location Beacon Database

Skyhook Wireless's patent, granted in 2008 to Jones *et al.*, describes a server that maintains a database of Wi-Fi access point locations across a metropolitan area [13]. Vehicles carrying a GPS receiver and a Wi-Fi radio drive nearly every street in a target area, and the server estimates each access point's position from many readings, discards readings that fall far from a weighted centroid, and updates the database as it discovers new access points. The patent differs from BevoTag in what it stores: it records where each access point is, whereas BevoTag records the signal pattern measured at each survey point inside a building. Google Patents lists the patent's adjusted expiration as September 30, 2026, so its outlier rejection and database updating are now free to adopt.

US 9,014,718: Location Sensing Using Channel Fading Fingerprinting

Intel's patent, granted in 2015 to Yang, Yang, and Guo, claims the core of a CSI fingerprinting system [14]. A device scans nearby access points, measures a "channel fading profile" from selected subcarriers, and finds its location by matching the profile to a database recorded during a survey, using a minimum-distance or maximum-likelihood rule. The dependent claims allow the database to sit on the device or a server, which mirrors BevoTag's open question of where to compute the position. This patent is the closest prior art to BevoTag's Wi-Fi stage, and it means that CSI fingerprint matching by itself is not patentable. Google Patents lists it as expired because its maintenance fee was not paid, so the method may be used freely.

US 9,632,169: Using Inertial Navigation to Calibrate an Indoor Positioning System

Ricoh's patent, granted in 2017 to Moraleda and Hull, addresses survey effort [15]. A surveyor walks between known points while inertial navigation tracks the path and Wi-Fi signatures are collected continuously; each signature is then assigned a position along a B-spline fitted to the path, and the surveyor may adjust the spline's control points on a display. This method could survey a floor far faster than stopping at each point. The patent remains in force until 2035, so the team will avoid the method of fitting and editing a spline path.

US 9,332,476: Correcting Indoor Positioning Using Wi-Fi Handovers

BlackBerry's patent, granted in 2016 to Shatsky and Kouznetsov, uses Wi-Fi to correct dead-reckoning drift when a device hands over to a new access point [16]. If the dead-reckoning position lies outside a "handover spot" around the new access point, the method of claim 1 resets the position to the nearest point on that spot's border. This patent confirms that drift correction from Wi-Fi is a recognized need, but it corrects only at handovers and only coarsely. BevoTag will instead correct the inertial estimate at every fingerprint fix with a filter, the approach that Evennou and Marx published in 2006 [11], and will not reset the estimate to a handover boundary. The patent remains in force until 2034.

US 9,479,920: Power Management in a Crowdsourced Lost-and-Found Service

Finally, Apple's patent, granted in 2016 to Mayor, describes the crowdsourced model behind tags such as the AirTag [17]. A Bluetooth Low Energy tag broadcasts an identifier; low-power subsystems in a passing phone record that identifier and the Wi-Fi access points the phone hears, each with a time stamp; and the phone's main processor periodically wakes, assigns the tag the stored location of the access point heard at the same time, and reports it to a server. The patent locates a tag only as precisely as the nearest access point and only when a participating phone is nearby, which are the two weaknesses that BevoTag is designed to overcome. Because BevoTag scans and reports by itself and uses no helper phones, it does not practice these claims, but the patent's strategy of batching measurements to keep the main processor asleep applies directly to BevoTag's battery target.

Table 2 summarizes the five patents and their status: the two expired patents are free to use, and BevoTag's design avoids the claims of the three active ones.

Table 2. Patents Relevant to BevoTag

Patent	Assignee	Relevance to BevoTag	Status (Google Patents)
US 7,414,988 [13]	Skyhook Wireless	Access point database with outlier rejection and updating	Expired Sept. 2026
US 9,014,718 [14]	Intel	CSI fingerprint matching against a survey database	Expired (unpaid fee)
US 9,632,169 [15]	Ricoh	Inertially tracked walking survey fitted with a B-spline	Active to 2035; avoid
US 9,332,476 [16]	BlackBerry	Drift reset to a handover-spot border	Active to 2034; avoid
US 9,479,920 [17]	Apple	Crowdsourced Bluetooth tag located by phones	Active to 2035; not practiced

IMPACT OF PRIOR-ART SEARCH ON DESIGN DECISION-MAKING

The search answered each of the four research questions and changed three parts of the design plan: the treatment of CSI, the survey protocol, and the fusion design. The following subsections answer each question in turn and state the decision that follows.

Wi-Fi Measurement: CSI Is Promising but Unproven on BevoTag's Hardware

The literature supports CSI's accuracy but leaves its feasibility on BevoTag in doubt: DeepFi's 30 to 39% error reduction [8] came from hardware far more capable than BevoTag's single-antenna tag, which will capture far fewer CSI frames whenever it listens passively [10].

The team therefore decided to make RSSI fingerprinting the baseline for every later design and to evaluate CSI as an addition to it. The first data logger experiment will record CSI and RSSI together in EER and measure how many usable CSI frames per second the tag captures from campus access points, both while associated with the campus utexas-iot network and while listening passively. If CSI improves room identification on held-out survey points by a margin comparable to DeepFi's, the design will keep it; otherwise, the design will drop it to save

processing and power. Because Intel's CSI fingerprinting patent has expired [14], either outcome is free of patent risk.

Survey Effort: Orientation and Time Matter More than Density

RADAR and the competition together show that a survey's coverage of conditions matters more than its density. RADAR reached nearly full accuracy with about 40 points, but its median error rose 67% when the survey and the user faced opposite directions [7], and the competition's leading Wi-Fi systems gained up to 0.94 m of error when furniture moved [12].

The team has accordingly revised the survey protocol in four ways:

1. Scans will be recorded facing at least four directions at each survey point.
2. Each benchmark area will be surveyed at three times of day, including class changes, when hallways are crowded.
3. The survey points from one visit will be held out for testing, so that accuracy figures reflect a survey that has aged.
4. The database will be kept current without full resurveys by adapting the outlier rejection and incremental updating of Skyhook's now-expired patent [13].

Sensor Fusion: A Map-Aware Filter Bounds Drift

Evennou and Marx's results show that fusion and floor-plan constraints together roughly halve fingerprinting error, from 3.32 m to 1.53 m [11], which suggests that BevoTag's 5 m median target is conservative and that room identification is the harder goal. Their study, the competition's mix of Wi-Fi and inertial entries [12], and BlackBerry's patent [16] all treat Wi-Fi as the source that corrects inertial drift, confirming the architecture in the Project Definition.

The fusion design has therefore narrowed to two candidate filters. An extended Kalman filter [18] is light enough to run on the tag but cannot easily use walls, whereas a particle filter can discard paths through walls using the floor plans already in the survey application but is computationally expensive [11]. Because of this trade-off, a particle filter would most likely run on the server,

and the tag would then send measurements rather than positions. The team will compare the two filters on logged EER walks before deciding.

Freedom to Operate and Novelty: The Combination Is the Contribution

The search shows that each of BevoTag's building blocks is already public: the patents in Table 2 claim most of them, and the research literature has published Wi-Fi and inertial fusion since at least 2006 [11]. No source found in the search, however, combines CSI-capable fingerprinting and inertial fusion on a self-contained, battery-powered tag that reports its own position without helper phones or new infrastructure. BevoTag will therefore build on the two expired patents and published methods, avoid the specific claims of the three active patents, and rest any claim of novelty on the integrated tag and its performance on a changing campus rather than on a new positioning algorithm.

CONCLUSION

This report presented a prior-art search that answered four open questions about BevoTag's Wi-Fi measurement, survey, sensor fusion, and patent landscape. The publications show that CSI can reduce error but that the tag will capture it less often, that a survey needs to cover several orientations and times of day, and that a map-aware fusion filter can roughly halve position error; the patents show that two have expired and that the design avoids the claims of the other three. RSSI fingerprinting will therefore serve as the baseline, CSI as an experiment, and the integrated, self-reporting tag as the project's contribution. Next, the team will measure CSI capture in EER, survey a benchmark area under the revised protocol, and compare two fusion filters to decide where BevoTag computes its position.

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